

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read, so on a large population it constrains nothing. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows

consistency, not cortical realization. The conditions yield a disconfirmable comparison: decoders fitted independently on overlapping regions should agree more on reported than on matched unreported stimuli at equal phase coherence. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012; Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent decoding evidence.

Binding-by-synchrony and communication-through-coherence accounts

17 treat phase relations as the medium of integration or of selective routing
 18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
 19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
 20 leave open what fact about synchronized activity makes it one content rather
 21 than coordinated activity carrying several. The contribution here is to sep-
 22 arate the three requirements and prove which follow from which. Phase
 23 order constrains shared content only through encoders related on overlaps,
 24 with a bound that grows as the square root of the population it is read
 25 on. Compatible local descriptions, not synchrony, determine a unique global
 26 state. A contracting self-map fixes one state but can reconstruct accurately
 27 only a family of bounded diameter. Each result names a measurement that
 28 phase data alone cannot supply.

29 The familiar noisy mean-field oscillator supplies a stationary coherence
 30 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
 31 condition for a declared actuator, bounds that connect phase order to con-
 32 tent only through specified encoders, and conditions under which compatible
 33 descriptions glue and an accurate reconstruction fixes the glued state. A
 34 Lean 4 development checks these implications and a joint witness satisfies
 35 their eight connecting hypotheses in a specified model. It proves neither that
 36 cortex satisfies those hypotheses nor that the glued state is experienced. The
 37 latter identification motivates the framework and needs its own endpoint.

38 The empirical question is which measurements could support or discon-
 39 firm the proposed connections. A phase resultant can miss a locally or-

40 dered winding field; stationary phase distributions can leave the coupling
 41 mechanism unidentified; and coherent adaptation can lose alignment with
 42 a designated content template. Independent decoders must identify shared
 43 quantities before their agreement can be tested. We therefore distinguish sta-
 44 tionary branches from finite observed onset, and compare prespecified field
 45 and connectivity onset models on held-out emergence recordings. The com-
 46 parison tests a specified spatial prediction, not the presence of experience.
 47 Section 10.2 states the comparison that bears on conscious content directly.
 48 Sections 2–5 develop the three requirements; Section 6 applies their mea-
 49 surement obligations to a digital candidate. Section 7 states the resource and
 50 composition premises, and Sections 8–9 examine observables and the cortical
 51 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
 52 gives their derivations and controls.

53 **2 Limits of phase order**

54 Several limits hold without the cortical hypothesis. Equal phases can coexist
 55 with incompatible local descriptions; phase order constrains shared content
 56 only through encoders related on overlaps. A winding field can have zero
 57 global resultant while remaining locally ordered. A constant readout can
 58 have a unique fixed point without reconstructing its input. A region that
 59 does not distinguish relevant states cannot encode them, and a site cannot
 60 report a change outside its causal past by a declared deadline. These lim-

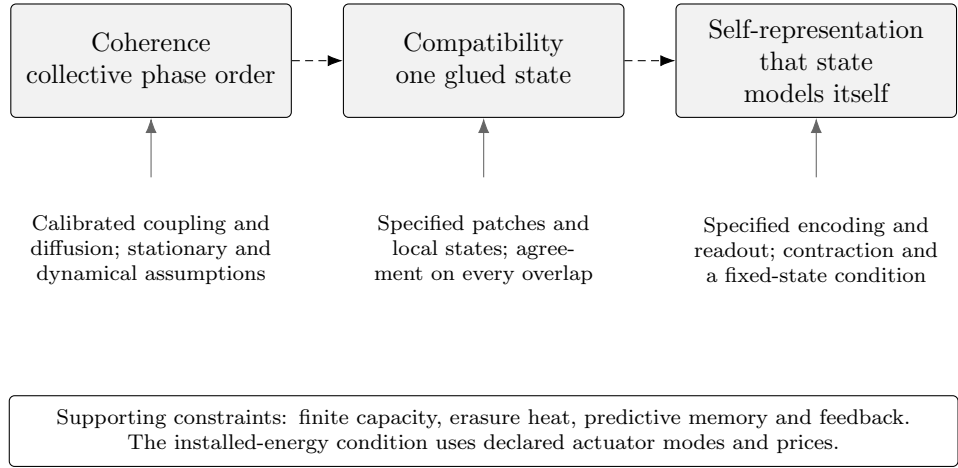


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

61 its reflect familiar phase oscillator and information constraints (Acebrón et
 62 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
 63 The sections below state each one beside the condition or measurement it
 64 motivates.

65 3 Coherence: physical coupling and its re- 66 source condition

67 3.1 Dynamics and the stationary threshold

68 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

69 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
70 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
71 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
72 quency heterogeneity and diffusion are distinct parameters. The stationary
73 calculation below uses identical frequencies and this scalar coupling. Ap-
74 plying it to a spatial kernel requires an aggregation rule and a check of the
75 resulting approximation.

76 For identical frequencies in their rotating frame, the positive normalized
77 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

78 where I_n are modified Bessel functions and ψ is the collective phase. The for-
79 mal derivation uses the absence of stationary current for a periodic gradient
80 drift to obtain this density, then verifies normalization and self-consistency.

81 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 82 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 83 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

84 Figure 2 shows this stationary branch.

85 Linearization about the uniform density gives rate $K/2 - D$ for the first
 86 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 87 exactly above threshold. About the coherent density, the quadratic form
 88 is strictly dissipative transverse to collective rotation, whose rate is zero.
 89 These linear estimates concern the limiting equation. Relating them to finite
 90 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 91 tion; exchange symmetry alone does not establish the needed independence
 92 (Sznitman, 1991).

93 **3.2 Installed energy bounds mean coupling**

94 A spatial model replaces the finite sum by integration against a kernel. One
 95 declared implementation carries response modes with continuous profiles ϕ_i ,
 96 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

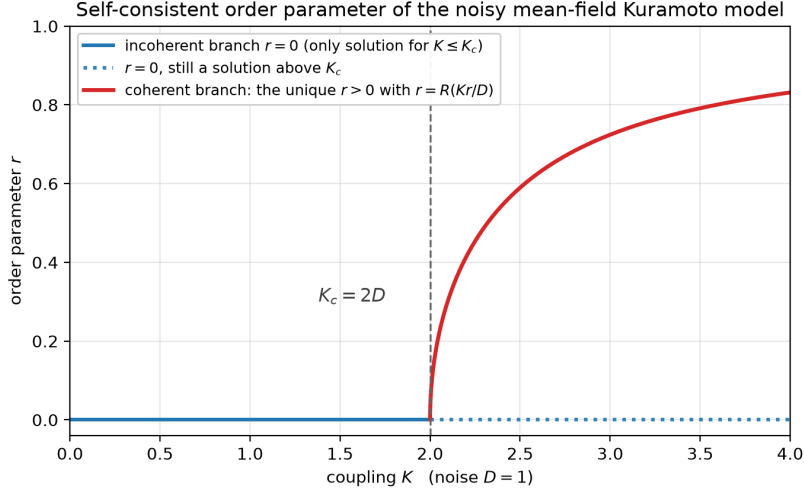


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

97 The same configuration sets the reciprocal kernel and stored energy. An
 98 executed transition changes the kernel through its changed occupancies and
 99 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 100 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 101 pair reconstruction on refining spatial partitions converges to its product-
 102 measure integral under the specified compactness and sampling assumptions;
 103 the refinement index is independent of the number of control updates.

104 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 105 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

106 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

107 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 108 coherent stationary branch and any growing Fourier mode about the uniform
 109 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 110 is necessary: energy above it need not produce sufficient coupling because
 111 the bound can be loose.

112 The theorem concerns stored energy in the declared actuator modes. The
 113 estimated cortical signalling expenditure and extracellular electrostatic en-
 114 ergy in Supplemental Material measure different quantities; neither identifies
 115 the occupancies, mode prices or their relation to metabolic supply. Conse-
 116 quently the bound supplies a necessary condition for a declared actuator,
 117 but no cortical value of κ or demonstration that cortex meets it. Applying
 118 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 119 a justified reduction.

120 4 Compatibility: from coordinated activity 121 to a global description

122 4.1 Why common timing is insufficient

123 For overlapping patches U and V , compatibility means that restricting either
124 local description to $U \cap V$ gives the same object. Descriptions can agree on
125 shared quantities while contributing different information elsewhere. Com-
126 patible sections of the specified sheaf have a unique global extension. On
127 a finite cover, local finite spatial measures agreeing as measures on every
128 overlap also have a unique finite-measure extension whose sheaf image is the
129 glued section. No mass-one normalization is imposed.

130 In the cup example, visual and reaching descriptions must express a
131 shared quantity, such as cup location, in common coordinates. Restriction
132 retains that quantity while dropping details particular to each description.
133 Agreement permits surface appearance and hand configuration to contribute
134 to one situation. The mathematical counterexample assigns the same phase
135 to every patch while two local mass profiles disagree on a shared site. Those
136 profiles have no common extension. Because they are chosen independently
137 of phase, synchrony places no constraint on their disagreement.

138 This distinction limits the biological interpretation. Receptive territories
139 and content domains must be related to spatial patches, and the descriptions
140 must support the required extension. Matching overlap marginals of de-

141 coded probability laws need not determine a unique global joint distribution
 142 (Abramsky & Brandenburger, 2011).

143 The cover is itself a choice, and the construction constrains it in one way
 144 only: synchronized patches agree where they overlap. Where they do not
 145 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 146 satisfies that obligation for *every* assignment of local descriptions and phases
 147 and the gluing theorem still returns a unique global section, which is the
 148 tuple it was handed. Membership of the class then reports nothing about the
 149 system beyond the patches covering it. What a gluing claim asks of a cover
 150 is therefore a non-degeneracy the theorem does not assume: that distinct
 151 patches meet, and that what they share is a quantity fixed independently of
 152 the agreement it is being used to establish.

153 4.2 When coherence does constrain content

154 A declared relation between phase and content gives a positive result. Let
 155 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 156 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 157 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 158 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 159 sites; their equality on overlaps is a supplied relation between descriptions of
 160 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 161 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 162 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.

163 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (6)$$

164 Here discrepancy is the norm of the difference, and r and N are read on
165 whichever population the bound is applied to. That choice is what decides
166 whether the estimate says anything. A patch carries an order parameter
167 of its own, and reading the bound there carries that patch's resultant and
168 that patch's site count, so a large population observed through small patches
169 is constrained by the small count. Read instead on the whole array, the
170 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
171 from an average over N sites. The order cannot be improved in general:
172 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
173 observations agree exactly at either scale. A shared scalar encoder is a special
174 case. Dropping the overlap relation permits disagreement even at perfect
175 locking. A common nonexpansive linear projection preserves agreement, but
176 agreement of a projection does not establish agreement of the full vector.

177 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
178 constrains anything only where the right-hand side falls below that. At
179 the sheet's own patch order, the bound is informative for at most 10 sites;
180 beyond that, the coherence-to-content link requires the bridge assumption
181 (E78) of the cortical hypothesis. Supplemental Material provides the detailed
182 derivation of this saturation, and states three conditional derivations that

183 replace the uniform factor $\sqrt{N}$ by a count of disagreeing pairs, a chain across
 184 a cover's shared sites, or a spectral gap at a frequency-locked configuration.

185 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 186 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 187 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 188 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

189 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 190 duce an initial mismatch. Approximate observations leave a nondecaying
 191 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 192 update to the phase process.

193 4.3 What approximate agreement determines

194 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 195 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 196 Weights that vanish outside their patches and sum to one at each site select
 197 a global profile by weighted averaging. The selection lies within ε of each
 198 patch there, and any two such selections differ by at most ε , independently
 199 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 200 $L\varepsilon$ across selections.

201 At zero tolerance, the selection is the unique exact extension. At every
 202 positive tolerance there are families with no exact extension; selection then

203 gives an approximation whose dependence on weights is bounded. Supple-
 204 mental Material gives a three-site witness attaining the bound. The mass-
 205 profile construction uses the scalar case of the content dynamics and addi-
 206 tionally requires nonnegativity of the initial profiles and observations. The
 207 general vector result supplies no identification with the spatial-measure sheaf.
 208 These constructions concern declared objects: a specified cover, contents
 209 valued in a normed space, and encoders related on overlaps. They select
 210 neither the neural cover nor the decoder, and identifying their contents with
 211 perceptual content, or establishing that the identification bears on experi-
 212 ence, is a further modelling task; Section 8 sets out what a measurement of
 213 it would have to establish.

214 **5 Self-representation: a state that models it-** 215 **self**

216 **5.1 The reflexive construction**

217 Compatibility gives a global state. Reflexivity relates it to an encoding within
 218 that state. Equip global sections with a complete metric. If the additional
 219 restriction-resonance condition identifies encoding with restriction to a region
 220 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

221 The region and readout are fixed across the state space. If F contracts and
 222 fixes the section s_* formed from the local descriptions, Banach's theorem
 223 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
 224 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
 225 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
 226 less than one for a relaxation time $\tau > 0$. Constructing a map with that
 227 factor remains a modelling obligation. A fixed contraction can reconstruct
 228 accurately only a restricted family of states: if its error is at most ε through-
 229 out that family, any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

230 In the cup example, the global state includes the represented scene and the
 231 perceiver's relation to it. A local encoding must retain enough information
 232 for the readout to recover that modelled state; cup position alone would be
 233 insufficient if hand configuration and viewpoint could not be reconstructed.

234 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 235 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

236 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 237 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 238 weakens.

239 5.2 Reconstruction across a family of states

240 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 241 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
 242 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

243 Thus M relevant states pairwise more than 2ε apart require at least M
 244 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 245 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 246 existing self-prediction map.

247 This is an additional accuracy criterion on a family, with no requirement
 248 for multiple exact fixed points. For the same fixed contracting map used by
 249 E89, however, the diameter bound above is an additional restriction: at zero
 250 error the family contains at most one metric-distinct state. A broad family
 251 therefore requires a different, possibly input-conditioned reconstruction map
 252 with its own accuracy and stability assumptions. A blind encoder or constant
 253 readout fails the stated criterion for separated states despite having a unique
 254 fixed point.

255 Where the encoding is read as the state's own restriction to a region, the
 256 criterion becomes a condition on the region. Restriction resonance requires
 257 the encoder and that restriction to identify the same pairs of states, so a
 258 boundary reconstructing two relevant states more than 2ε apart while its

259 region reads them identically is not resonant. Read forwards, a resonant
 260 boundary that reconstructs a pairwise separated family restricts injectively
 261 on it. Which states a region reads identically is fixed by where the region
 262 sits, before an encoding is chosen, so the resolution the family demands is
 263 the region's and no readout supplies it. The implication runs one way: a
 264 region that does resolve the family is not thereby shown to carry a resonant
 265 encoding, and E89 supplies no such resolution.

266 Resonance and resolution are jointly satisfiable without a local fold. Take
 267 the region to be the whole substrate and use the identity readout: restric-
 268 tion is the identity, so this boundary is resonant and reconstructs every de-
 269 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 270 E89's strict contraction premise on a nontrivial metric space. A properness
 271 condition on the region is therefore a separate requirement, and the fixed-
 272 contraction diameter bound remains in force when E89 is also required. With
 273 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 274 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 275 so the family is no larger than the δ -packing number of what the region car-
 276 ries. That is the shape a continuous family of states needs, and at the whole
 277 substrate it says nothing, because the region's range is everything. The gain
 278 L is declared and derived here for no physical readout, and at unbounded
 279 gain the bound says nothing.

280 **5.3 Agreement after a change takes communication**

281 Snapshot compatibility does not specify how quickly descriptions respond
282 when a shared quantity changes. In a synchronous network whose updates
283 read only the site’s own state and declared neighbours’ messages, two initial
284 configurations agreeing on a site’s causal past through T rounds produce the
285 same state there at round T . The deadline consequence is the one Section 2
286 states, and it follows from the reconstruction bound in Section 5.2: a report
287 from that site is a readout of a code that the two interventions leave identical.

288 A three-site delay line witnesses failure after one round and exact report-
289 ing after two; cutting the path prevents delivery at every round. A report
290 may still happen to be correct about one fixed world. The bound concerns
291 responsiveness across interventions.

292 Throughout this section *prediction* names a reconstruction map, and the
293 relevant family of states, the metric, the tolerance, the communication graph
294 and the deadline are all declared inputs, as are the represented variables and
295 the fidelity counted as accurate. Reading the map as temporal neural pre-
296 diction would require dynamics; reading the resulting code count as physical
297 memory capacity would require a hardware model, which a state-dependent
298 readout would have to extend.

299 6 What the conditions constrain in a digital 300 candidate

301 The conditions bear on a large language model (LLM) running on graphics
302 processing units (GPUs) only through measurements, and only for a declared
303 task. For a finite spatial coupling operator with eigenvalues $|\lambda_1| \geq \dots \geq |\lambda_N|$,
304 any linear encoder and readout factoring through d real coordinates pays re-
305 construction error at least $\sum_{i>d} \lambda_i^2$. That rank premise is a property of an
306 explicitly linear bottleneck; query/key dimension does not supply it for soft-
307 max attention. On one synthetic two-view task whose target is the glued
308 state and a self-relative coordinate, a rank-two linear decoder has predicted
309 training-error floor 2.000 and held-out error 2.000 on a parity-held-out family.
310 The deadline premise is similarly specific: within one forward pass, the causal
311 mask confines position t to its causal past, so an intervention outside it can-
312 not be reconstructed there by that step. These results rank no architecture
313 family, imply no general inferiority of GPU hardware, and settle no question
314 of machine experience; a claim of unity or self-representation for such a sys-
315 tem commits its maker to identifying the shared quantities, the code and its
316 separation of the declared family under intervention. Supplemental Material
317 gives the bottleneck derivation, the task controls, the information ceiling of
318 token outputs and the hardware scope.

319 7 Resource premises and composition

320 Two thermodynamic statements enter, with different standing. The first is
321 proved for the phase model: for a smooth positive periodic density with prob-
322 ability current J and diffusion D , the current functional $\sigma_J = \int J^2/(D\rho) d\theta$
323 bounds how fast order can change, $\int_0^\tau \sigma_J dt \geq [\arcsin r(\tau) - \arcsin r(0)]^2/(D\tau)$
324 while $0 < r < 1$, and stationary order has no positive floor. The second is
325 conditional: the installed-energy condition of Section 3.2 is necessary for its
326 declared actuator, and restricts a physical candidate only once mode prices,
327 coupling reduction and diffusion are measured. Neither is a heat or power
328 statement until the current is identified with a physical flux, a reservoir at a
329 stated temperature is specified and local detailed balance holds for the steps
330 concerned; only then does σ_J convert to entropy production and the finite
331 register budgets to Landauer heat (Landauer, 1961; Norton, 2005; Shenker,
332 2000). No composition edge derives coherence, compatibility or reconstruc-
333 tion from a heat bound. Finite-register capacity obeys $H(p) \leq \log |X|$, and
334 a separate boundary result concerns a field reaching disconnected vacuum
335 components (Kibble, 1976); Table 1 states the connecting assumptions. Sup-
336 plemental Material gives the current-cost derivation, predictive-memory and
337 feedback balances, learning and repeated-operation budgets, and the sensi-
338 tivity of the metabolic comparison to its assumptions.

339 7.1 What the composition establishes

340 The passive and feedback composition theorems each accept eight connecting
341 implications (Table 1). In the active branch, a named substep at the register’s
342 thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (11)$$

343 with B the register’s heat. The upper allocation follows from a ledger whose
344 operations deliver that total heat to one reservoir and whose other opera-
345 tions do not increase joint entropy. Those ledger assumptions are physical
346 inputs. Neither Landauer’s lower bound nor microscopic reversibility sup-
347 plies them. Finite two-bit witnesses exhibit positive forward and reverse
348 elementary paths, local detailed balance at one thermal scale, and interac-
349 tion energies of order $k_B T$; these conditions alone do not quantify proximity
350 to equilibrium. Cortex has not been shown to satisfy these premises. The
351 illustrative cortical metabolic expenditure over one region and one phase-
352 diffusion time is 2.4×10^{14} times kT at body temperature, but identifying it
353 with the register heat budget B requires a common physical ledger. Read-
354 ing the register at the scale where its premises are physical, such as a single
355 channel, requires E45Active to carry that step’s budget to the convergence
356 of a continuum kernel. The active branch is an alternative route to the same
357 node rather than a stronger one, and no result of Section 3.2 passes through
358 it: the installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical	A finite-capacity system realizes a field
<i>capacity and</i>	premise	leaving its vacuum manifold. Capacity and
<i>boundary</i>		boundary are independent roots.
E23	Bridge	The boundary is associated with a finite
<i>boundary to</i>	assumption	register whose update leaves states
<i>register</i>		unreachable.
E34 /	Bridge	Passive: predictive memory at the register’s
E34Active	assumption	thermal scale and heat budget, with
<i>register to</i>		nonzero waste. Active: positive paths, local
<i>dissipation</i>		detailed balance and allocation of register
		heat to the named substep; the allocation
		uses the ledger identity and other
		operations’ compressiveness. Both are
		exhibited at thermal-scale heats on finite
		state spaces with positive reverse paths,
		which is not the regime a cortical register
		occupies.
E45 /	Modelling	Convergent spatial coupling energies.
E45Active	assumption	Declared mode hardware supplies a kernel
<i>couplings to</i>		limit for either branch. A learning witness
<i>continuum</i>		computes installation and work from its
		own executed law. Convergence follows
		from mesh refinement rather than from the
		budget, which selects no regime at
		macroscopic scale. Refinement and physical
		identifications remain inputs.
E56	Physical	A specified physical field realizes the
<i>field</i>	commitment	limiting coupling and satisfies the kernel
<i>identification</i>		conditions.
E67	Physical	That field operates above the
<i>above</i>	commitment	synchronization threshold.
<i>threshold</i>		

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

359 Both branches also take a thermodynamic cover carrying exact over-
360 lap compatibility and a reflexive boundary specifying encoding and read-
361 out. They conclude that the glued section is the unique fixed point of self-

362 prediction. Compatibility and accuracy at that section enter as premises;
363 uniqueness follows. Identifying encoding with restriction needs the separate
364 resonance condition of Eq. (8).

365 Each of the eight edges is supplied rather than derived from the one above
366 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
367 pendent roots rather than a step from one to the other; both E45 discharges
368 take their convergence from refinement data and not from the thermody-
369 namic bound; E78 states that coherence relaxes the supplied cover, and the
370 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
371 coders already shared on overlaps. The inferential content of the composition
372 is its last step, where contraction and a fixed glued section give uniqueness.
373 What composing them establishes is that the premises are consistent, named
374 in one place and machine-checked together, and not that coherence follows
375 from installed energy or self-representation from coherence.

376 Joint witnesses combine a register, boundary, refining mesh and small
377 spatial cover; active witnesses add controlled thermal updates and a priced
378 installation. They establish satisfiability under their stated identifications.
379 Mesh convergence follows from refinement data, and the witnesses establish
380 no common cortical mechanism. Supplemental Material maps the claims to
381 their Lean declarations and records the counterexamples. The development
382 declares no additional axioms beyond its libraries' foundational axioms.

383 8 What the observations establish

384 The composition holds under its premises. Whether observation can estab-
385 lish those premises is a separate question, and the three requirements it
386 connects concern different objects: phase dynamics, local descriptions and
387 internal encodings. Their observables must retain these distinctions. The
388 following controls show which mechanisms remain indistinguishable, which
389 temporal approximations fail, which states one observable cannot separate,
390 and what independent content measurements must establish. Each is a finite
391 numerical regime on a declared model, with full parameters and controls in
392 Supplemental Material, and none of them calibrates a cortical quantity.

393 8.1 Stationary distributions leave the mechanism un- 394 determined

395 Equation (2) gives a concentration–coherence relation with no fitted shape
396 parameter. Estimate r from circular resultant length and a from density
397 shape: inverting R from the measured r would make agreement automatic.
398 Even independently estimated agreement identifies a statistical family shared
399 by different mechanisms.

400 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
401 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
402 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
403 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

404 and diffusion D_0 have that same density. Their stationary distributions co-
 405 incide window by window despite different causes.

406 Resolved dynamics supply additional information. In the ideal model
 407 with continuous latent paths, quadratic variation accumulates at rate $2D$
 408 and conditional increment means constrain drift; a filtered observation of
 409 the phase needs its own observation model before either is read off. Separat-
 410 ing endogenous coupling from a matched common drive additionally requires
 411 a measured drive or controlled perturbation; identifying electromagnetic cou-
 412 pling requires evidence specific to that interaction. Supplemental Material
 413 gives the equivalence construction and its numerical check.

414 8.2 Finite ramps change the observable onset

415 The square-root law in Eq. (3) describes stationary solutions. A changing
 416 coupling requires time for the phase distribution to approach those solutions.
 417 In a deterministic mean-field control that crosses the threshold symmetrically,
 418 none of the scanned rates meets the declared tracking tolerance: slowing
 419 the crossing does not remove the terminal residual under that initialization.
 420 A slow control variable therefore supplies no general guarantee of branch
 421 tracking at onset. Supplemental Material gives the residual, rate range and
 422 admissible rates away from threshold.

423 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 424 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 425 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay

426 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 427 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 428 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 429 onset exponents differ from those of the exact stationary branch evaluated
 430 on the same grids. At the fastest speed the ensemble mean never reaches the
 431 fit window, so its fitted slope does not estimate stationary onset.

432 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 433 makes the apparent exponent depend on ramp speed and fitting window.
 434 The specified deviation-based critical ramp speed is not bracketed by this
 435 sweep. With fluctuating coupling, the relevant stationary comparison also
 436 depends on rate: the mean coupling approximates order in the tested fast
 437 regime, while averaging stationary order over coupling works for the tested
 438 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 439 neither approximation.

440 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 441 $v^{0.444}$. The interval from fitting those means is [0.394, 0.494]; resampling
 442 replicas independently within each speed gives a 95% interval [0.398, 0.496],
 443 with fraction 0.019 of 10000 bootstrap fits at or above 1/2. This fraction
 444 describes the resampling distribution under the fitted protocol; it is not a
 445 test of the ideal exponent. A deterministic mean-field run seeded at the
 446 fluctuation floor gives exponent 0.425 with the same estimator and escape
 447 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls

448 show that the criterion affects the fitted exponent, but do not prove that it
 449 accounts for the entire finite-population shortfall. The larger-population low-
 450 threshold fit has substantial log residuals; matched-speed fits do not converge
 451 toward one-half across the tested sizes and criteria. Some trajectories also
 452 cross the absolute escape level before K_c , so the first postcritical crossing can
 453 yield a zero or grid-limited delay. This finite sweep establishes no common
 454 asymptotic exponent.

455 The ideal delayed-onset calculation motivates a square-root scaling hy-
 456 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 457 tify the coupling medium. A within-subject emergence design can vary the
 458 propofol taper and test how an operational onset changes with that manipu-
 459 lation, but interpreting the result as a coupling-rate law requires a measured
 460 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 461 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 462 describes how D could be estimated from resolved phase traces.

463 Additional controls address integration bias, population size and spatial
 464 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 465 $[1.958, 2.009]$ contains 2.

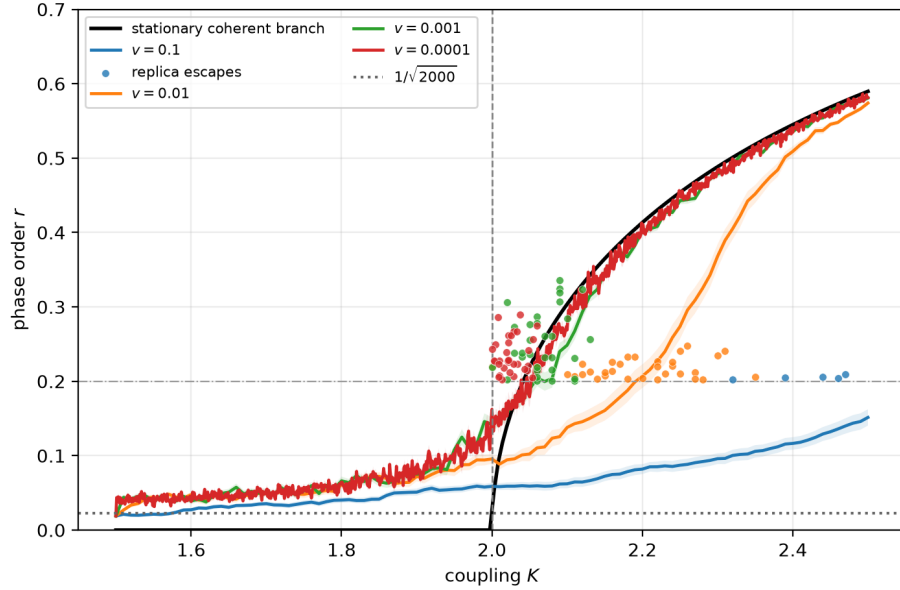


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica’s first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

466 8.3 A winding state leaves the order parameter am- 467 biguous

468 The global order parameter r discards spatial arrangement. On a periodic
469 sheet, a phase field can wind once around the sheet and have $r = 0$ while
470 neighbouring sites remain locked. At identical frequencies, every such wind-
471 ing is stationary for an isotropic symmetric kernel: contributions at opposite
472 separations cancel. It violates the quarter-turn initial-spread premise of the

473 convergence result, which is why that result can still select the uniform state.

474 In the spatial control, the winding persists below the interpolated de-
 475 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
 476 and winding starts under the same coupling, diffusion and kernel end at
 477 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 478 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 479 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 480 realizations. These are finite controls, not a cortical wave identification or a
 481 derived transition.

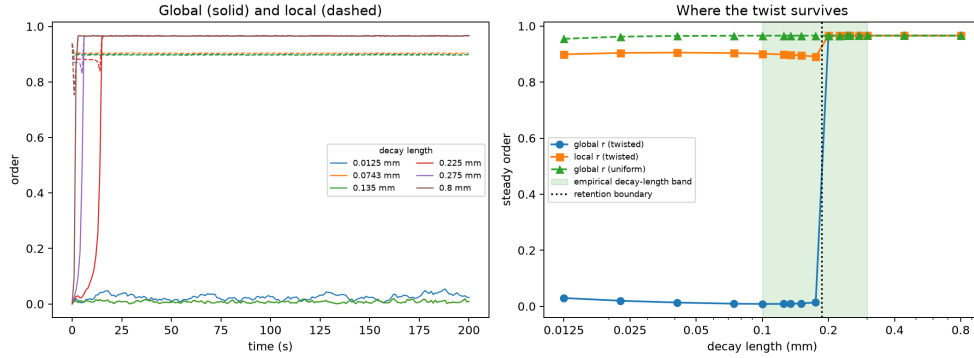


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

482 Linearization about a winding reweights each separation’s coupling by

483 the cosine of its phase advance. The resulting symmetric kernel has explicit
 484 mode rates: nonnegative reweighted coupling gives no growing mode, while
 485 a positive rate destabilizes that winding. The criterion concerns linear rates,
 486 not asymptotic stability, and the measured retention boundary need not
 487 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 488 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 489 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 490 winding from that model even though the phase-only equations retain it as
 491 a stationary state.

492 Neither model turns these states into measured cortical spirals. The pe-
 493 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 494 no propagation speed, and the amplitude variable has no calibrated relation
 495 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 496 and defect results. Locally, Eq. (6) can still bound content disagreement on a
 497 winding patch when shared encoders are supplied, although its global $r = 0$
 498 form is uninformative. A nonzero ring winding number q also obstructs a
 499 global real-valued phase lift; it has no identified content meaning here. Thus
 500 an onset criterion based only on r can register a change of basin as well
 501 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 502 observable.

503 8.4 Coherent adaptation can lose environmental align- 504 ment

505 An adaptive coupling rule might supply representational structure, but its
506 objective must reward the relevant agreement. For symmetric coupling, pair-
507 wise sine terms cancel in the population mean drift. Minimizing squared drift
508 at that fixed mean therefore rewards equalizing drift across nodes. In the
509 frequency-cluster control, this favors coupling between unlike-frequency clus-
510 ters, whereas the diagnostic environmental template rewards coupling within
511 clusters.

512 The formalization proves descent of the squared-drift objective for a spec-
513 ified gradient flow at fixed phases. Its functional differs from thermodynamic
514 entropy production: at a coherent stationary density the probability current
515 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
516 rial gives the operator, its gradient and the conditions of the descent result.

517 The finite control co-evolves noisy phases and symmetric, nonnegative
518 couplings at fixed total resource. Its template is a diagnostic covariance
519 of latent signals, never a training input. Across three illustrative seeds,
520 second-half order remains between 0.9167 and 0.9222, while the within-
521 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
522 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
523 trols are reported in Supplemental Material.

524 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

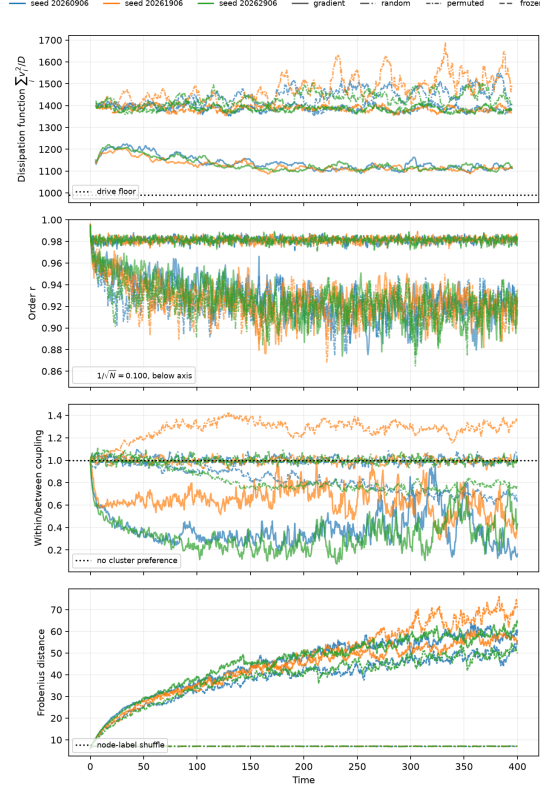


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

547 tent, phase carries a share 1.000 of the content entropy when content is built
548 from it and 0.000 when it is not. A phase-derived statistic reports com-
549 patibility on phase-locked patches whose local descriptions disagree. The
550 implemented content statistic detects their incompatibility. These controls
551 validate a measurement on known constructions; a neural content model and
552 decoder remain to be established.

553 Choosing the cover also chooses a candidate subject: trivial descriptions
554 can agree while carrying little content, and a partition can conceal disagree-
555 ment. The useful choice must therefore be constrained by what neural ac-
556 tivity encodes. The exact composition assumes compatibility of its supplied
557 cover. The content mechanisms above identify additional hypotheses under
558 which agreement can arise; they leave the empirical choice of content and
559 the connection to that cover explicit.

560 **8.6 Estimator calibration on exploratory EEG**

561 The concentration range and the dependence structure decide whether the
562 concentration-coherence relation can be tested at all. Under the specified
563 log-density estimator and bias-floor criterion, discrimination at $a = 0.542$
564 requires 3000 independent sites in the scanned designs, rising to 31000 when
565 sites share a mean in groups of 10. These counts describe that estimator and
566 dependence model at a declared design concentration; the intended recording
567 requires its own calibration. Supplemental Material reports the bin-count
568 controls and the full dependence-aware design.

569 The estimators were also exercised on public propofol-sedation EEG (Ba-
 570 jwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels,
 571 band-passed, Hilbert-transformed and pooled over pairs and 100-ms win-
 572 dows. Across 8 subjects the pooled concentration spans 0.087–0.142, and
 573 the residual from the Bessel relation is small. That agreement is uninforma-
 574 tive. Pooling over a window does not remove the common rotation of the
 575 phases, so a perfectly phase-locked 10-Hz signal passed through the same
 576 pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its
 577 instantaneous spatial resultant is one. The bipolar montage also removes
 578 spatial coherence: a field whose raw channels have instantaneous resultant
 579 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform
 580 pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel
 581 ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks
 582 the estimator’s behaviour and establishes no spatial concentration, coher-
 583 ence level or test of the stationary density. Sedation recordings also leave the
 584 awakening hypothesis untested.

585 An informative dataset therefore needs an estimand that removes the
 586 common rotation, a montage whose effect on spatial phase is modelled, suf-
 587 ficient concentration range or precision, calibrated estimators and an ob-
 588 servation model for dependence and source mixing. Intracranial recordings
 589 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
 590 informative concentration range, but that range must be measured. The
 591 design counts above are calculations for the specified estimator, not sam-

592 ple requirements transferable to another recording. Public mouse calcium
593 imaging provides thousands of spatially located neurons across waking, sleep
594 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
595 phase-estimation band used here. Site count, state coverage and bandwidth
596 are separate requirements.

597 **9 The cortical hypothesis and an awakening** 598 **test**

599 The cortical proposal identifies a physical source of coupling and asks whether
600 its changes help explain recovery. The tests in Section 8 determine what
601 observations must distinguish before a phase trace can support this identifi-
602 cation.

603 **9.1 Physical identification and calibration**

604 The cortical hypothesis identifies part of effective coupling with endogenous
605 extracellular electromagnetic interactions. Fields can modulate spike timing
606 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
607 propose coordination of distributed neural information (Pinotsis & Miller,
608 2022; McFadden, 2020). Phase organization under propofol motivates study-
609 ing this regime (Bastos et al., 2021), but does not measure coupling.

610 The formal field predicate requires a jointly continuous kernel, a normal-

611 ized substrate, coupling equal to the kernel average, and no single site pro-
612 viding more than half that strength. Toy constructions satisfy it. A cortical
613 realization must specify the scale at which the kernel and its reduction ap-
614 ply: calcium imaging reports different state-dependent modularity patterns
615 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

616 Cortical columns are the obvious candidate unit, and they answer one
617 of the two demands the formalism makes. As the oscillator index they are
618 a serious proposal: they are delineated by response properties and anatomy
619 rather than by the agreement they would be used to establish (Mountcastle,
620 1997), and a phase can be estimated from a column-sized territory at the
621 electrode spacings used for cortical wave measurements (Townsend et al.,
622 2015). As the content cover they are a poor one, for three separate reasons.
623 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
624 the compatibility obligation without constraining anything (Section 4). The
625 cover condition additionally requires the patches to exhaust the substrate,
626 while much of association cortex has no accepted columnar parcellation. And
627 the column’s standing as a canonical cortical unit is contested (Horton &
628 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
629 supplies the oscillator index; the content cover remains an open empirical
630 question, which the bridge assumption E78 records.

631 The spatial reduction is also testable. A row-sum aggregation at zero
632 diagonal tracks the uniform-coupling control over the tested spatial decay
633 range, while frequency heterogeneity displaces the operational threshold. The

634 fit window biases both estimates relative to $2D$; agreement with the control
 635 is therefore the supported conclusion. Supplemental Material reports the
 636 reduction, its numerical error and the distinct assumptions needed for the
 637 kernel-average identification in Eq. (5).

638 The field commitment requires physical evidence beyond a choice of math-
 639 ematical representation, and the available support comparisons do not supply
 640 it. A finite matrix spread over positive-mass cells preserves its total cou-
 641 pling and phase correlation exactly, while point-supported vanishing under
 642 an atomless measure and a disadvantage from excluding a better-correlated
 643 pair concern specified support constraints; Section 6 reads the same compar-
 644 isons against a digital candidate. Neither supplies energy prices.

645 A spike-time shift alone supplies no coupling rate. The dimensionless
 646 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 647 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (12)$$

648 This rate describes how induced timing shifts accumulate into drift. Nei-
 649 ther it nor the installation factor κ follows from a heat bound. Testing the
 650 physical threshold requires independent estimates of the interaction kernel,
 651 its aggregation and diffusion in the same cortical regime.

9.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (12) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives a deterministic tracking approximation that fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (13)$$

This is a deterministic tracking approximation from Eq. (3) at fixed D ; the crossing controls in Section 8.2 show that it fails as an observed-onset approximation under the tested initialization. A separation of geometry and phase time scales can support tracking away from threshold. The awakening

672 hypothesis needs a calibrated dynamical prediction for an operational onset
673 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
674 recovery latency.

675 **9.3 A protocol for the awakening experiment**

676 Neural inertia describes a behavioural barrier and hysteresis between anaes-
677 thetic induction and emergence (Friedman et al., 2010); recovery can pass
678 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
679 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
680 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
681 mechanism from a synaptic one. A possible discriminator is spatial onset
682 relative to independently estimated local field coupling and network connec-
683 tivity. The present mean-field model has no spatially varying κ and derives
684 no nucleation order, so a field-based onset map needs a separate model spec-
685 ified before the observations are inspected.

686 For that comparison, partition the recording into prespecified regions i
687 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
688 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
689 and a connectivity score c_i from a separately estimated structural network.
690 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
691 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

692 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (14)$$

693 Predicted onset is the first crossing of a region-specific threshold fixed from
694 pre-onset fluctuations. Fit the two scales on training sessions, then com-
695 pare held-out onset times and site order under the same observation model.
696 Larger field scores predict earlier onset only when baseline order and diffu-
697 sion do not reverse the ranking. This declared approximation omits interre-
698 gional coupling and saturation; a successful ranking would support its field
699 covariate, not prove ephaptic causation. Supplemental Material specifies the
700 controls and failure conditions.

701 A within-subject protocol can manipulate emergence rate, with spatially
702 resolved recordings to compare the proposed field and connectivity models:

- 703 1. **Manipulate emergence rate.** Vary the propofol infusion taper
704 across sessions within the same subject, recording drug concentrations
705 or a validated pharmacokinetic proxy to quantify the manipulation.
- 706 2. **Measure behavioural recovery.** Record time-to-response to a stan-
707 dard stimulus as the independent behavioural endpoint, rather than
708 estimating ΔK directly.
- 709 3. **Resolve spatial onset.** Identify anatomical locations where coher-
710 ence first crosses the baseline-scaled threshold and score the two maps

711 of Eq. (14) on held-out sessions. The two covariate maps are fixed
712 before the outcome sessions are inspected.

713 **4. Match physical controls.** Confirm that the observed coherence-to-
714 amplitude relation holds under the specific local diffusion D and ob-
715 servation noise. The model’s quadratic-variation rate $2D$ applies to
716 the latent phase; a band-limited analytic phase is smooth below its
717 correlation time, and on a constructed field its squared-increment rate
718 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
719 fore requires an explicit observation model for filtering and sampling.

720 Predeclare the recording window, regional fluctuation thresholds, and a
721 shared timing-error model. Fit the two scales and that observation model
722 on training sessions only; on held-out sessions, score the observed site-onset
723 times and sites that never cross by the same regional predictive rule. Report
724 site order as a secondary outcome and behavioural time-to-response sepa-
725 rately. Sessions without a crossing are censored observations, not zero delays
726 or exclusions. Compare the two scores only where independently estimated
727 field and connectivity maps predict distinguishable onset patterns at the
728 recording resolution. A held-out field-model advantage would support that
729 specified spatial predictor; it would not identify an ephaptic cause, shared
730 content or experience. Supplemental Material gives the controls and pilot
731 quantities needed to judge whether the comparison is informative.

732 **10 Discussion**

733 **10.1 Unity and a minimal self as interpretive motiva-** 734 **tion**

735 We propose the state glued from compatible local descriptions as a formal
736 correlate of conscious unity, and its accurate internal reconstruction as a cor-
737 relate of a minimal self in the sense of a pre-reflective point of view on the
738 scene (Gallagher, 2000). In the cup example, this would mean one description
739 that preserves the cup’s relation to the perceiver as well as the scene. Com-
740 patibility permits distinct contents and uncertainty; the shared quantities
741 must be chosen from independent evidence about what each region encodes.
742 The reconstruction need not be a verbal or autobiographical account. Its
743 proposed relation to experienced perspective remains an interpretation. Nei-
744 ther identification is among the eight hypotheses in Table 1, and satisfying
745 them would not measure experience.

746 **10.2 A comparison that could disconfirm the link to** 747 **content**

748 The framework predicts that conscious content tracks compatibility, not co-
749 herence. Present matched stimuli that are reported on some trials and not
750 on others, for example at masking or rivalry threshold, and fit decoders for a
751 shared stimulus quantity independently on two overlapping cortical regions.

752 Compare decoded agreement on the overlap between reported and unreported
753 trials after matching their phase coherence. The proposed link predicts higher
754 agreement on reported trials at equal coherence. Equal agreement, or agree-
755 ment that follows coherence and not report, counts against it. A self-model
756 comparison has the same form: reconstruction of perceiver-relative variables,
757 such as hand position relative to the target, from a declared local encoding
758 should be more accurate on reported trials. These are correlational tests of
759 the proposed correlates; report is itself an access measure, and a no-report
760 design would be needed to separate content from access.

761 **10.3 What a test of the full proposal must establish**

762 A test needs separate estimates of effective coupling and diffusion, shared
763 content on overlapping regions, and reconstruction from a specified encoding
764 at a declared fidelity. The cover, relevant state family and decoder must
765 be fixed without selecting them for agreement. An independent experiential
766 endpoint is needed to evaluate the interpretation; absence of a report alone
767 does not define absence of experience. A unified episode with incompati-
768 ble decoded descriptions would challenge the unity identification. A unified
769 episode lacking the proposed self-encoding would challenge the minimal-self
770 claim. Conversely, satisfying the mathematical conditions without such an
771 episode would challenge their sufficiency as correlates.

772 For the physical route, cortical actuator modes and prices, the geometry-
773 coupling relation and a dynamical onset model remain to be measured. The

774 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
775 diction. Controlled changes in coordination and independently decoded con-
776 tent agreement would address the representational connection. The synthetic
777 decoder exercise calibrates part of these measurements, and the EEG exercise
778 identifies what a pooled bipolar estimand cannot measure; neither identifies
779 a neural content code or tests experience.

780 **10.4 Explanatory scope**

781 Integrated-information, global-workspace and predictive-processing accounts
782 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
783 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
784 relations among coupling, local content and reconstruction. Its formal field
785 condition requires a continuous kernel, a normalized substrate, a specified
786 average coupling and noise, and a site-contribution bound. Those conditions
787 do not identify an electromagnetic substrate; the extracellular interpretation
788 needs the separate calibration of Section 9. The joint witness shows that
789 the formal hypotheses can hold together in a declared model. A cortical
790 realization and an experiential interpretation require additional evidence.

791 Three limitations bound every claim above. Synchrony is not itself an
792 experiential measure, and nothing here establishes a consciousness criterion:
793 the conditions are proposed correlates to be tested against report or another
794 independent endpoint. The present EEG analysis checks a measurement
795 pipeline and does not validate any experiential claim. The formal results are

796 conditional on declared encoders, covers and maps whose cortical counter-
797 parts are unmeasured.

798 The remaining explanatory question is whether these relations occur in
799 neural activity and matter for experience. The framework identifies candidate
800 correlates and tests of their physical and representational premises. It does
801 not explain why satisfying them would be accompanied by experience.

802 11 Conclusions

803 Phase coherence, compatibility and self-reconstruction are separable condi-
804 tions. Coherence constrains shared content only through encoders shared
805 on overlaps and with a bound that grows as the square root of the popula-
806 tion read; compatible local descriptions, not synchrony, determine one global
807 state; and a contracting self-map fixes that state while reconstructing only a
808 family of bounded diameter. Each result replaces an assumption about syn-
809 chrony by a measurement: independently decoded agreement on overlaps,
810 a declared encoding and its fidelity, and calibrated coupling and diffusion.
811 The comparison of Section 10.2 states how such measurements could count
812 against the proposed link to conscious content.

813 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (11)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (12)	Field-to-coupling conversion rate, timing sensitivity and frequency

Table 2 (continued)

Symbol	Defined	Meaning
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover’s nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (12)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

814 Data and code availability

815 The Lean development, simulation scripts, tests and saved numerical
816 summaries are maintained in the GitHub repository `bobaseb/coupling_`
817 `gated_cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and
818 Mathlib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publi-
819 cation simulation values are generated from saved summaries without re-

820 running production sweeps. The exploratory EEG analysis uses Open-
821 Neuro dataset ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022,
822 1033, 1045, 1056, resting awake and sedation runs 1–2); preprocessing
823 and selection are described in Supplemental Material and implemented in
824 `simulations/empirical_collapse.py`. The repository contains the repro-
825 ducible analysis and does not redistribute the raw recordings.

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829 **Use of AI tools**

830 AI assistance to this work was substantial and is not separable by task. Sev-
831 eral coding assistants and chat interfaces were used in an interleaved way
832 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
833 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
834 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
835 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
836 accessed with the Hermes CLI. Between them they contributed to the Lean
837 development, the simulation and analysis code, and the drafting and revision
838 of both documents. Because the tools were used in combination and in alter-

839 nation, no section, proof, figure or result is attributable to any one of them,
840 and none is claimed to be. The author is responsible for all content, including
841 everything a tool produced. Machine checking establishes the stated Lean
842 results under their hypotheses; it does not validate biological interpretations
843 or replace scientific review of generated code and prose.

844 **References**

- 845 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
846 electric fields emerge from and allow representational drift. *NeuroImage*,
847 253, 119058.
- 848 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
849 tion theory: from consciousness to its physical substrate. *Nature Reviews*
850 *Neuroscience*, 17(7), 450-461.
- 851 Singer, W., & Gray, C. M. (1995). Visual feature integration and the temporal
852 correlation hypothesis. *Annual Review of Neuroscience*, 18, 555-586.
- 853 Fries, P. (2015). Rhythms for cognition: communication through coherence.
854 *Neuron*, 88(1), 220-235.
- 855 Gallagher, S. (2000). Philosophical conceptions of the self: implications for
856 cognitive science. *Trends in Cognitive Sciences*, 4(1), 14-21.

- 857 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
858 Mind. *Oxford University Press*.
- 859 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
860 tational theory of conscious processing. *Current Opinion in Neurobiology*,
861 25, 76-84.
- 862 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
863 *Reviews Neuroscience*, 11(2), 127-138.
- 864 Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness:
865 A case for biological computationalism. *Neuroscience & Biobehavioral Re-*
866 *views*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.
- 867 Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial
868 consciousness through the lens of neuroscience. *Trends in Neurosciences*,
869 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.
- 870 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
871 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- 872 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
873 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
874 doi:10.1017/S0140525X26106906.
- 875 Gurnee, W., et al. (2026). Verbalizable Representations Form a
876 Global Workspace in Language Models. *Transformer Circuits Thread*.
877 arXiv:2607.15495.

- 878 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
879 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
880 40, 24-50.
- 881 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
882 parts of the free energy principle. *Entropy*, 23(3), 293.
- 883 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
884 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.
- 885 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
886 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.
- 887 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
888 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
889 *Progress in Neurobiology*, 226, 102465.
- 890 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
891 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
892 9(8), 45-60.
- 893 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
894 testable hypothesis about the characteristics of conscious as opposed to
895 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- 896 McFadden, J. (2020). Integrating information in the brain’s EM field: the
897 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
898 niaa016.

- 899 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
900 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 901 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
902 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
903 95-103.
- 904 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
905 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 906 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
907 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 908 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
909 *logical Reviews*, 98(1), 239-389.
- 910 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
911 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.
- 912 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
913 without a function. *Philosophical Transactions of the Royal Society B*,
914 360(1456), 837-862. doi:10.1098/rstb.2005.1623.
- 915 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National*
916 *Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.
- 917 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-

918 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
 919 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

920 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
 921 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
 922 ity of complexity measures to capture dreaming during propofol sedation.
 923 *Scientific Reports*, 15, 32746.

924 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
 925 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
 926 brain. *Science*, 342(6156), 373-377.

927 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
 928 *Physiological Reviews*, 88(4), 1277-1340.

929 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
 930 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

931 Logothetis, N. K., Kayser, C., & Oeltermann, A. (2007). In vivo measure-
 932 ment of cortical impedance spectrum in monkeys: implications for signal
 933 propagation. *Neuron*, 55(5), 809–823. doi:10.1016/j.neuron.2007.07.027.

934 Gabriel, S., Lau, R. W., & Gabriel, C. (1996). The dielectric properties of bio-
 935 logical tissues: II. Measurements in the frequency range 10 Hz to 20 GHz.
 936 *Physics in Medicine and Biology*, 41(11), 2251–2269. doi:10.1088/0031-
 937 9155/41/11/002.

938 Norton, J. D. (2005). Eaters of the lotus: Landauer’s principle and the return
939 of Maxwell’s demon. *Studies in History and Philosophy of Science Part B:*
940 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

941 Landauer, R. (1961). Irreversibility and heat generation in the computing
942 process. *IBM Journal of Research and Development*, 5(3), 183-191.

943 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
944 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

945 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
946 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

947 Strogatz, S. H., & Mirollo, R. E. (1991). Stability of incoherence in a popu-
948 lation of coupled oscillators. *Journal of Statistical Physics*, 63, 613–635.

949 Acebrón, J. A., Bonilla, L. L., Pérez Vicente, C. J., Ritort, F., &
950 Spigler, R. (2005). The Kuramoto model: A simple paradigm for
951 synchronization phenomena. *Reviews of Modern Physics*, 77, 137–185.
952 doi:10.1103/RevModPhys.77.137.

953 Ott, E., & Antonsen, T. M. (2008). Low dimensional behavior of
954 large systems of globally coupled oscillators. *Chaos*, 18, 037113.
955 doi:10.1063/1.2930766.

956 Dörfler, F., & Bullo, F. (2014). Synchronization in complex net-
957 works of phase oscillators: A survey. *Automatica*, 50, 1539–1564.
958 doi:10.1016/j.automatica.2014.04.012.

- 959 Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature*
960 *Neuroscience*, 20, 340–352. doi:10.1038/nn.4497.
- 961 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
962 of prediction. *Physical Review Letters*, 109(12), 120604.
- 963 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
964 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.
- 965 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
966 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
967 031015.
- 968 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
969 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
970 PhysRevLett.111.180603.
- 971 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
972 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 973 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
974 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
975 181.
- 976 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
977 archive.pitt.edu/115.

- 978 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
979 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
980 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 981 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
982 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
983 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 984 Dai Pra, P., & den Hollander, F. (1996). McKean–Vlasov limit for interacting
985 random processes in random media. *Journal of Statistical Physics*, 84, 735–
986 772. doi:10.1007/BF02179656.
- 987 Bertini, L., Giacomin, G., & Pakdaman, K. (2010). Dynamical aspects of
988 mean field plane rotators and the Kuramoto model. *Journal of Statistical*
989 *Physics*, 138, 270–290. doi:10.1007/s10955-009-9908-9.
- 990 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
991 bifurcations with additive noise. *Probability Theory and Related Fields*,
992 122, 341–388. doi:10.1007/s004400100174.
- 993 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
994 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
995 doi:10.1088/1367-2630/13/11/113036.
- 996 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
997 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

998 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
 999 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

1000 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
 1001 terns underlie complex brain dynamics and are related to cognitive pro-
 1002 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
 1003 023-01626-5.

1004 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
 1005 and dissipation with partially observed fluctuating signals. *Physical Review*
 1006 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

1007 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
 1008 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
 1009 cell resolution functional networks during unconsciousness are segre-
 1010 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
 1011 doi:10.1016/j.celrep.2025.116902.

1012 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
 1013 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
 1014 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.

1015 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the
 1016 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
 1017 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.

- 1018 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
1019 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.
- 1020 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.
- 1021 Baer, S. M., Erneux, T., & Rinzel, J. (1989). The slow passage through a
1022 Hopf bifurcation: delay, memory effects, and resonance. *SIAM Journal on*
1023 *Applied Mathematics*, 49(1), 55–71. doi:10.1137/0149003.
- 1024 Friedman, E. B., Sun, Y., Moore, J. T., Hung, H.-T., Meng, Q. C., Perera, P.,
1025 Joiner, W. J., Thomas, S. A., Eckenhoff, R. G., Sehgal, A., & Kelz, M. B.
1026 (2010). A conserved behavioral state barrier impedes transitions between
1027 anesthetic-induced unconsciousness and wakefulness: evidence for neural
1028 inertia. *PLoS ONE*, 5(7), e11903. doi:10.1371/journal.pone.0011903.
- 1029 Hudson, A. E., Calderon, D. P., Pfaff, D. W., & Proekt, A. (2014). Recovery
1030 of consciousness is mediated by a network of discrete metastable activity
1031 states. *Proceedings of the National Academy of Sciences*, 111(25), 9283–
1032 9288. doi:10.1073/pnas.1408296111.
- 1033 Proekt, A., & Hudson, A. E. (2018). A stochastic basis for neural inertia
1034 in emergence from general anaesthesia. *British Journal of Anaesthesia*,
1035 121(1), 86–94. doi:10.1016/j.bja.2018.02.035.